Formal Requirements Document (FRD)

Project: University Infrastructure Project Management System (UIPMS)
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Document Control

Version	Date	Author	Summary of Changes
0.1	10 September 2025	Maladji TE	Initial draft

1. Introduction

1.1 Purpose

This document defines the requirements for the University Infrastructure Project Management System (UIPMS). It describes what the system must do, who will use it, how it should behave, and the rules it must follow—especially for project registration, deliverable tracking, communication, invoicing, approvals, payments, reporting, archiving, and closure.

1.2 Background

University infrastructure projects involve collaboration between internal staff (project managers, facilities leadership, finance) and external service providers. The current process is fragmented across emails, spreadsheets, and separate systems. UIPMS will centralize this into a single, auditable, and integrated system aligned with the university's standard business procedures.

1.4 Definitions, Acronyms, Abbreviations

- **UIPMS:** University Infrastructure Project Management System
- **PM:** Project Manager
- **Director of Facilities (Director):** Approver after PM recommendation
- **Finance Responsible Official (Finance):** Final approver for payment processing
- **Service Provider (SP):** External vendor/contractor
- **Deliverable:** A tangible output (document, build stage, inspection certificate, etc.) linked to payments
- **Milestone/Stage:** A defined step in project progress; may bundle multiple deliverables
- **ITS:** University's enterprise finance/ERP system used for payment execution and recording (exact vendor/system TBD)

2. Stakeholders & Users

Role	Description	Key Responsibilities in UIPMS
Project Manager (PM)	University staff managing projects	Create projects, define deliverables/milestones, review SP

Role	Description	Key Responsibilities in UIPMS
		updates, recommend invoices, monitor progress
Service Provider (SP)	External vendor executing work	Submit deliverable updates, upload diagrams/designs, submit invoices, respond to comments
Director of Facilities	Senior approver	Review PM recommendations, approve/reject invoices
Finance Responsible Official	Payment authority	Review approved invoices; process/trigger payment in ITS
Facilities/Project Admin	Operational support	Data quality checks, user provisioning, reporting
System Administrator (IT)	Technical operations	User management, configuration, integration, support
Executives/Auditors	Oversight	Read-only access to reports/audits

3. User Stories

1. **As a Project Manager**, I want to register and track projects with deliverables, so that I can manage timelines and validate reports for payment.
2. **As a Service Provider**, I want to upload deliverables and submit invoices, so I can get paid when agreed-upon stages are reached.
3. **As the Director**, I want to approve recommended invoices, so that I uphold governance and compliance.
4. **As a Finance Official**, I want to receive pre-approved invoices, so I can process payments accurately.

4. Functional Requirements

4.1 User & Access Management

- The system must provide secure authentication for staff (SSO if available) and SPs (username password; MFA configurable).

- Role-based access control (RBAC) with least privilege: PM, SP, Director, Finance, Admin, read-only.
- SPs must only see projects to which they are explicitly assigned.
- Account lifecycle: invite, activate, suspend, deactivate; password reset.

4.2 Project Registry & Setup

- Create/edit/view projects with mandatory fields: Project ID, Title, Description, Department, PM, SP(s), Start Date, Expected End Date, Budget (optional), Location/campus (optional).
- Define milestones/stages with sequence number, name, description, target date, and payment allocation (% or fixed amount).
- Define deliverables per milestone with description, acceptance criteria, due date, attachments (optional).
- Upload, version, and preview architecture diagrams/designs (PDF, PNG, JPG, DWG/DXF if feasible).
- Project statuses: Draft → Active → On Hold → Completed → Closed; with transitions controlled by role and validation.

4.3 Activities & Work Tracking

- PM/SP can add activities/work items linked to deliverables, with start/due dates, assignee (optional), status, and comments.
- Bulk import of deliverables/activities via CSV template (optional).

4.4 Communication & Correspondence

- Project-level discussion threads with message posting, reply, @mentions, and file attachments.
- All correspondence stored and searchable by project (includes timestamps, author, attachments).
- Email notifications configurable for mentions, status changes, approvals, and due/overdue reminders.

4.5 Status Updates & PM Validation

- SP can submit status updates on deliverables/milestones (progress %, evidence attachments, comments).
- PM must validate and endorse SP updates before they affect milestone completion status.

- PM can request changes/clarifications, system logs revision requests and responses.
- Only PM-endorsed completion marks a deliverable/milestone as “Complete (Endorsed)”.

4.6 Invoicing & Approval Workflow

- SP submits invoices through UIPMS: invoice number, date, amount, currency (if applicable), tax/VAT fields, line items linked to deliverables/milestones, and attachments (PDF required).
- The system must automatically check invoice alignment with agreed milestones/deliverables and their endorsed completion status.
- If an invoice is submitted **before** the agreed stage is reached, the system must **issue an alert** stating payment should only occur once the stage is reached.
    - **(Configurable):** Institution may configure alert-only vs. hard block vs. escalation-required override with justification.
- Workflow routing:
    - PM reviews invoice and provides a **Recommendation** (Recommend Approve / Recommend Reject) with comments.
    - System forwards to **Director of Facilities** for **Approval/Reject**.
    - If approved, system forwards to **Finance Responsible Official** for payment processing.
- Rejection must include a reason. The invoice then returns to SP/PM for correction/resubmission.
- All actions time-stamped with actor, action, and comments in an immutable audit log.
- The system must support the university’s standard business procedures (SOPs) and capture any required reference numbers or forms.

4.7 Payment Handling & ITS Integration

- The system must maintain a payment schedule per project derived from milestones (percentages or fixed amounts).
- Upon payment completion in ITS, UIPMS must update the invoice’s payment status with **amount paid** and **payment date** via integration.

- Support integration patterns: secure file exchange (SFTP), or message queue; retries and error handling with alerts.
- Display payment status history and reconciliation notes.

4.8 Reporting & Dashboards

- PM dashboard: active projects, milestone completion %, upcoming deadlines (next 30/60/90 days), overdue deliverables, invoices awaiting action, spend vs. budget (if provided).
- Director & Finance dashboards: invoices pending approval, approval aging, exceptions (early invoices), payments in progress.
- Historical project archive: searchable by year, department, SP, status, location.
- Export reports to CSV/PDF; scheduled report emails.

4.9 Project Closure & Archiving

- A project can be **Closed** only when: all deliverables are endorsed as complete, all invoices are paid or cancelled, and closure checklist is completed (documents finalized).
- Closure generates a final summary (PDF) including key data, approvals, and attachments index.
- Closed projects moved to read-only archive per retention policy.

4.10 Audit, Logging, & Search

- Full audit trail for create/update/delete actions, approvals, communications, and file versions.
- Global search across projects, deliverables, invoices, and messages; with filters by date, status, role.

5. Data Model (Conceptual)

Conceptual Data Model Explanation

The system's data model defines the main entities (data objects) and how they relate to each other. Below is an explanation of each entity and its purpose:

1. Project

Represents an infrastructure project being managed.
Key attributes include:

- **ProjectID:** Unique identifier for the project
- **Title & Description:** Name and details of the project
- **Department:** Department responsible for the project
- **PM:** Assigned Project Manager
- **SPs []:** List of associated Service Providers
- **StartDate & ExpectedEndDate:** Project timeline
- **Status:** Current state (e.g., Draft, Active, Completed)
- **Budget & Location:** Optional financial and site details

2. Milestone (or Stage)

A major checkpoint in the project that groups deliverables and often ties to payment.
Attributes:

- **MilestoneID:** Unique identifier
- **ProjectID:** Links the milestone to its project
- **Name & SeqNo:** Name and sequence order
- **TargetDate:** Planned completion date
- **PaymentAllocation:** Percentage or amount of total payment linked to this milestone
- **Status:** Current progress (e.g., Not Started, In Progress, Complete)

3. Deliverable

A specific, tangible output required to complete a milestone.
Attributes:

- **DeliverableID:** Unique identifier
- **MilestoneID:** Links deliverable to its milestone
- **Name & Description:** What the deliverable is
- **DueDate:** Deadline for completion

- **AcceptanceCriteria:** Conditions for approval
- **Status:** Current state (e.g., Submitted, Endorsed)
- **Evidence []:** Supporting documents or proof

4. Activity (or Work Item)

Represents smaller tasks linked to a deliverable.
Attributes:

- **ActivityID:** Unique identifier
- **DeliverableID:** Links activity to a deliverable
- **Title & Assignee:** Task name and responsible person
- **StartDate & DueDate:** Task timeline
- **Status & Comments:** Progress and notes

5. Document

Stores files like architecture diagrams or designs.
Attributes:

- **DocumentID:** Unique identifier
- **LinkedTo:** Indicates if the document belongs to a Project, Milestone, or Deliverable
- **Type:** Diagram, design, or other
- **Version & FileMeta:** Version control and file details

6. Invoice

Represents a billing request from a service provider.
Attributes:

- **InvoiceID:** Unique identifier
- **ProjectID & SupplierID:** Links invoice to project and vendor
- **Date, Amount, Currency, Tax/VAT:** Financial details
- **LineItems[]:** Breakdown of charges linked to milestones/deliverables

- **Status:** Draft, Submitted, Recommended, Approved, Rejected, Paid

7. Invoice Line Item

Detailed entry within an invoice.
Attributes:

- **LineItemID:** Unique identifier
- **InvoiceID:** Parent invoice
- **LinkedDeliverable/Milestone:** What the charge relates to
- **Amount & Description:** Cost and explanation

8. User

Represents system users (staff or service providers).
Attributes:

- **UserID:** Unique identifier
- **Role(s):** PM, SP, Director, Finance, etc.
- **Name, Email, Org:** Contact and organization details
- **Status:** Active, Suspended, etc.

9. Approval

Tracks decisions in workflows (e.g., invoice approval).
Attributes:

- **ApprovalID:** Unique identifier
- **Context:** What is being approved (e.g., Invoice)
- **Actor & Action:** Who approved/rejected and what action was taken
- **Timestamp & Comments:** When and why

10. Payment

Stores payment confirmation details from the finance system.
Attributes:

- **PaymentID:** Unique identifier
- **InvoiceID:** Linked invoice
- **AmountPaid & PaymentDate:** Actual payment details
- **ITSRef:** Reference from the finance system
- **Status:** Paid or Pending

11. Communication

Captures all correspondence related to a project.
Attributes:

- **MessageID:** Unique identifier
- **ProjectID:** Linked project
- **Author & Body:** Who sent the message and its content
- **Attachments[]:** Files shared in the message
- **Timestamp & ThreadID:** When and conversation grouping

Relationships

- A **Project** can have multiple **Milestones**, and each Milestone can have multiple **Deliverables**.
- Each **Deliverable** can have multiple **Activities**.
- A **Project** can have multiple **Invoices**, and each Invoice can have multiple **Line Items** linked to Deliverables or Milestones.
- A **Project** can have multiple **Documents**, which may also link to Milestones or Deliverables.
- Each **Invoice** can have multiple **Approvals** and one optional **Payment** record.

6. Workflow Specifications

6.1 Project Registration

Preconditions: PM authenticated; SP exists or is invited.
Main Flow:

1. PM creates project with required fields.
2. PM defines milestones and payment allocations.
3. PM defines deliverables per milestone and due dates.
4. PM uploads initial diagrams/designs.
5. Project moves from **Draft** to **Active** (validations applied).
   Postconditions: Project visible on dashboards; SP notified if assigned.

6.2 SP Status Update & PM Endorsement

1. SP logs a status update (progress %, comments, evidence).
2. System flags updated deliverables/milestones as **"Awaiting PM Endorsement."**
3. PM reviews, requests changes if needed, or endorses.
4. On endorsement, deliverable/milestone status becomes **Complete (Endorsed)** and can contribute to payment eligibility.

6.3 Invoice Submission & Approvals

1. SP submits invoice, linking line items to endorsed deliverables/milestones.
2. System validates alignment and triggers **early-invoice alert** if applicable (configurable behaviour).
3. PM provides **Recommendation** (approve/reject with comments).
4. Director **Approves/Rejects** (if reject → back to SP/PM).
5. If approved, Finance processes payment via ITS.

6.4 Payment Status Update via ITS

1. ITS marks payment completed.
2. UIPMS receives update (API/file), records **Amount Paid** and **Payment Date** and sets invoice status to **Paid**.
3. Dashboards and reports reflect updated financials.

6.5 Project Closure

1. PM runs closure checklist (all endorsed, invoices paid/cancelled).
2. System generates closure summary.
3. Status moves to **Closed**; project archived read-only.

7. Non-Functional Requirements (NFRs)

- **(Usability):** Plain language labels, in-app help, and tooltips. Training guides and short videos available.
- **(Accessibility):** WCAG 2.1 AA compliance; keyboard navigation; alt text for images; colour contrast checks.
- **(Performance):**
  - 95% of pages load ≤ 2.5s under normal load (≤ 200 concurrent users).
  - Search returns initial results ≤ 3s for queries over 50k records.
- **(Availability):** 99.5% monthly uptime (excluding planned maintenance); maintenance windows announced 72 hours in advance.
- **(Scalability):** Support growth to 10x current project volume without redesign; horizontal scaling where feasible.
- **(Security):** TLS 1.2+ in transit; encryption at rest; RBA
- **(Interoperability):** Standards-based APIs); secure file transfer (SFTP) options; time-zone aware timestamps (UTC storage).
- **(Data Retention):** Archive closed projects per policy (e.g., retain 7–10 years); configurable purge after retention period with legal hold option.
- **(Logging & Monitoring):** Centralized logs, alerting on integration failures and workflow SLA breaches.

8. Security, Privacy & Compliance

- Comply with university InfoSec and **POPIA** (lawful processing, purpose limitation, minimal collection, secure storage, data subject rights).
- Least-privilege access; SPs constrained to specific projects.
- Data classification and handling for sensitive documents (drawings may be confidential).
- **Audit trail** must record who did what and when; exportable for internal/external audit.
- Incident response: defined contacts, severity levels, and notification timelines.

- Backups with periodic restore testing; disaster recovery RTO/RPO targets (e.g., RTO 8h, RPO 4h).

9. Reporting & Dashboards (Illustrative)

- **PM Dashboard KPIs:**
  - Projects by status (Active/On Hold/Completed/Closed)
  - Milestone completion % (burn-up chart)
  - Upcoming & overdue deliverables (list + counts)
  - Invoices: awaiting recommendation/approval, aging buckets
  - Spend vs. budget (if budget captured)
- **Director/Finance KPIs:**
  - Pending approvals (count, aging)
  - Exceptions (invoices submitted before stage completion)
  - Payments processed this month/quarter, average cycle time
- **Historical Reports:**
  - Project throughput by year/department
  - Vendor performance (on-time delivery, change requests)
  - Payment compliance vs. contractual stages

10. Integration (ITS)

- **Interfaces:**
  - **Outbound to ITS:** Approved invoice details (if needed by ITS) or reference to retrieve.
  - **Inbound from ITS:** Payment confirmation (invoice ID/reference, amount paid, payment date, ITS transaction ID).
- **Security:** Mutual TLS, IP allowlists, OAuth2 or signed tokens.
- **Resilience:** Retry with exponential backoff; dead-letter queue; monitoring and alerts.

- **Reconciliation:** Nightly job to verify invoice → payment consistency; discrepancy report to Finance.

11. Acceptance Criteria & Test Scenarios

11.1 Key Acceptance Criteria (Examples)

- Creating a project with required fields moves status to **Active** and becomes visible on PM dashboard.
- SP's status update does not mark milestone complete until PM endorsement is recorded.
- Submitting an invoice linked to an unendorsed milestone triggers an **alert** before routing for approval.
- Invoice approval path follows: PM Recommendation → Director Approval → Finance processing.
- After payment in ITS, UIPMS shows **Paid** with correct **Amount** and **Payment Date**.
- All messages and file uploads appear in project correspondence and audit log.
- A project cannot be **Closed** until all deliverables are endorsed and invoices are reconciled.

11.2 Sample Test Cases (Condensed)

1. **Early Invoice Alert**
   - *Given* Milestone M1 is not endorsed
   - *When* SP submits an invoice referencing M1
   - *Then* System shows an alert: "Payment should only be made once the agreed stage is reached."
2. **PM Endorsement Gate**
   - *Given* SP updated a deliverable to 100% with evidence
   - *When* PM endorses the deliverable
   - *Then* Milestone progress updates and eligibility for invoicing becomes **true**.
3. **Approval Routing**
   - *Given* PM recommends approve

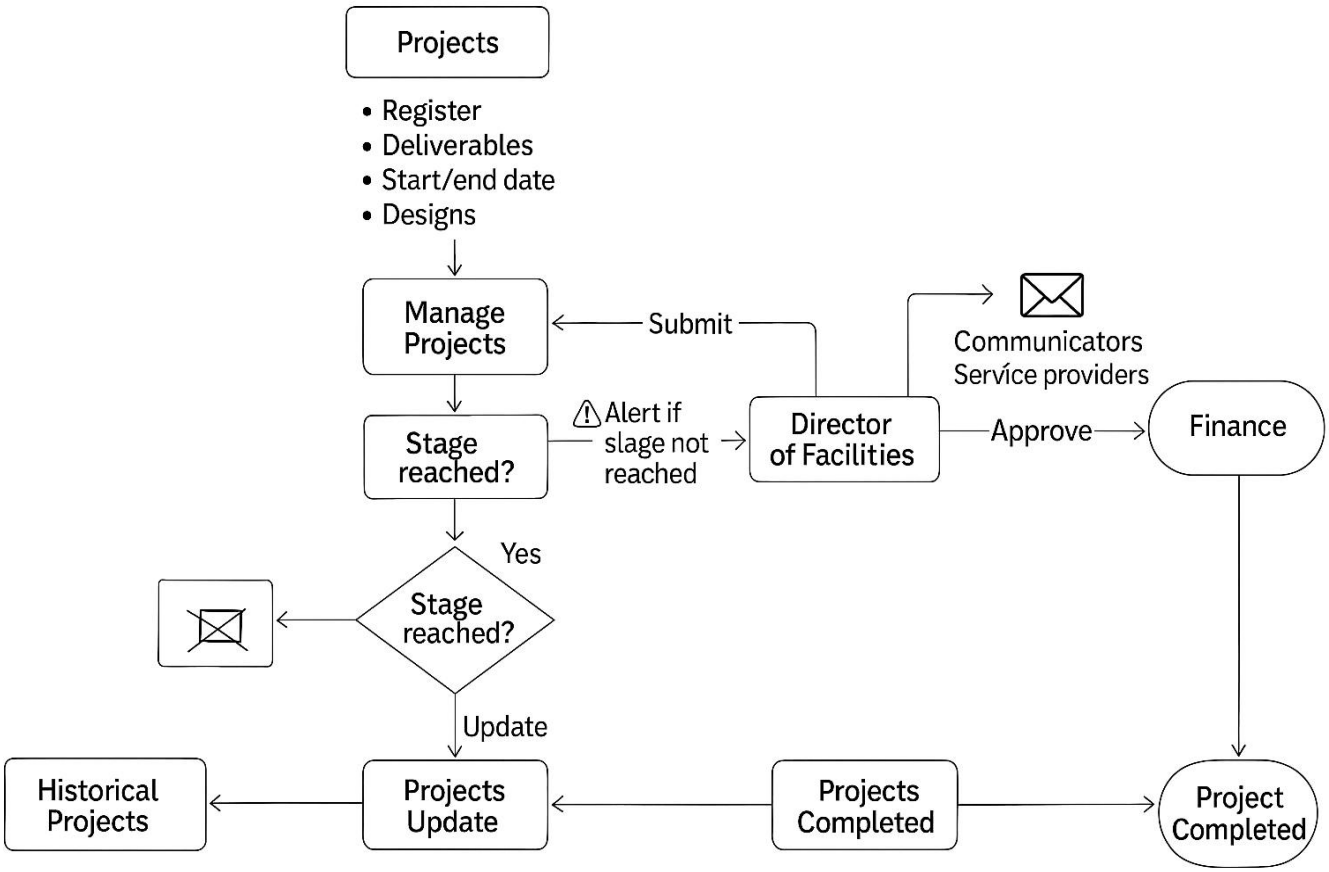
- *When* Director approves
- *Then* Invoice is forwarded to Finance and is visible in Finance queue.

4. **ITS Payment Update**
   - *Given* Invoice I1 is approved
   - *When* ITS posts payment status (amount/date)
   - *Then* Invoice I1 becomes **Paid**, dashboards update within 5 minutes.
5. **Security—SP Access**
   - *Given* SP belongs only to Project A
   - *When* SP tries to access Project B
   - *Then* Access is denied and logged.

12. Risks & Dependencies

- **Dependencies:** Availability of ITS integration endpoints; timely policy clarifications; user provisioning processes.
- **Key Risks & Mitigations:**
  - Incomplete SOP mapping → Early workshops with Finance/Facilities; configurable workflow engine.
  - SP adoption challenges → Simple UX, training, and phased onboarding.
  - Integration downtime → Retry queues, manual reconciliation fallback.
  - Data sensitivity (designs) → Access controls, watermark previews, legal disclaimers.

13. business workflow diagram


Projects
• Register
• Deliverables
• Start/end date
• Designs
Manage Projects
Submit
Communicators
Service providers
Stage reached?
Alert if slage not reached
Director of Facilities
Approve
Finance
Yes
Stage reached?
Update
Historical Projects
Projects Update
Projects Completed
Project Completed